Supplementary Material

Valid prediction intervals for transferred NIR calibrations: reusing the transfer standards with a leave-one-standard-out jackknife+

All tables and figures in this document were generated automatically from the machine-readable result files of the study (results/).

Table S1. Main results for every scenario

Table S1. Coverage of nominal 90% intervals and median interval score for every primary scenario (α = 0.10). CI: percentile bootstrap over repetitions.

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
corn	moisture	mp5	10	ASTM-PI	0.035 [0.032, 0.039]	4.71	True
corn	moisture	mp5	10	Gauss-LOO	0.897 [0.888, 0.907]	1.36	False
corn	moisture	mp5	10	SCP-S	0.063 [0.059, 0.068]	4.54	True
corn	moisture	mp5	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	moisture	mp5	10	TJ+	0.919 [0.909, 0.927]	1.42	False
corn	moisture	mp5	10	TJ+N	0.922 [0.912, 0.930]	1.48	False
corn	moisture	mp5	10	Transport	0.386 [0.361, 0.410]	3.06	True
corn	moisture	mp5	10	WCP-S	0.306 [0.292, 0.322]	∞	True
corn	moisture	mp5	15	ASTM-PI	0.037 [0.033, 0.042]	4.59	True
corn	moisture	mp5	15	Gauss-LOO	0.893 [0.883, 0.901]	1.29	False
corn	moisture	mp5	15	SCP-S	0.064 [0.059, 0.070]	4.39	True
corn	moisture	mp5	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	moisture	mp5	15	TJ+	0.942 [0.934, 0.949]	1.4	False

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Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
corn	moisture	mp5	15	TJ+N	0.934 [0.926, 0.941]	1.41	False
corn	moisture	mp5	15	Transport	0.332 [0.309, 0.354]	3.12	True
corn	moisture	mp5	15	WCP-S	0.220 [0.208, 0.233]	∞	True
corn	moisture	mp5	20	ASTM-PI	0.035 [0.030, 0.040]	4.32	True
corn	moisture	mp5	20	Gauss-LOO	0.907 [0.896, 0.917]	1.19	False
corn	moisture	mp5	20	SCP-S	0.068 [0.061, 0.076]	4.14	True
corn	moisture	mp5	20	SCP-T-split-N	0.914 [0.903, 0.924]	1.41	False
corn	moisture	mp5	20	TJ+	0.915 [0.905, 0.925]	1.22	False
corn	moisture	mp5	20	TJ+N	0.911 [0.902, 0.920]	1.26	False
corn	moisture	mp5	20	Transport	0.307 [0.286, 0.327]	2.99	True
corn	moisture	mp5	20	WCP-S	0.204 [0.190, 0.217]	∞	True
corn	moisture	mp6	10	ASTM-PI	0.026 [0.023, 0.029]	5.32	True
corn	moisture	mp6	10	Gauss-LOO	0.903 [0.893, 0.913]	1.52	False
corn	moisture	mp6	10	SCP-S	0.051 [0.047, 0.055]	5.12	True
corn	moisture	mp6	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	moisture	mp6	10	TJ+	0.919 [0.910, 0.929]	1.6	False
corn	moisture	mp6	10	TJ+N	0.918 [0.909, 0.927]	1.61	False
corn	moisture	mp6	10	Transport	0.366 [0.342, 0.392]	3.48	True
corn	moisture	mp6	10	WCP-S	0.303 [0.288, 0.318]	∞	True
corn	moisture	mp6	15	ASTM-PI	0.032 [0.028, 0.036]	5.12	True
corn	moisture	mp6	15	Gauss-	0.898	1.38	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
				LOO	[0.888, 0.907]		
corn	moisture	mp6	15	SCP-S	0.056 [0.051, 0.061]	4.94	True
corn	moisture	mp6	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	moisture	mp6	15	TJ+	0.941 [0.933, 0.949]	1.51	False
corn	moisture	mp6	15	TJ+N	0.935 [0.927, 0.942]	1.53	False
corn	moisture	mp6	15	Transport	0.313 [0.291, 0.335]	3.57	True
corn	moisture	mp6	15	WCP-S	0.223 [0.210, 0.237]	∞	True
corn	moisture	mp6	20	ASTM-PI	0.033 [0.028, 0.038]	4.83	True
corn	moisture	mp6	20	Gauss-LOO	0.906 [0.896, 0.916]	1.28	False
corn	moisture	mp6	20	SCP-S	0.058 [0.051, 0.065]	4.68	True
corn	moisture	mp6	20	SCP-T-split-N	0.912 [0.902, 0.922]	1.53	False
corn	moisture	mp6	20	TJ+	0.910 [0.900, 0.920]	1.28	False
corn	moisture	mp6	20	TJ+N	0.909 [0.899, 0.918]	1.38	False
corn	moisture	mp6	20	Transport	0.288 [0.267, 0.310]	3.37	True
corn	moisture	mp6	20	WCP-S	0.195 [0.181, 0.208]	∞	True
corn	oil	mp5	10	ASTM-PI	0.184 [0.175, 0.195]	1.35	True
corn	oil	mp5	10	Gauss-LOO	0.911 [0.903, 0.920]	0.597	False
corn	oil	mp5	10	SCP-S	0.453 [0.438, 0.467]	0.874	True
corn	oil	mp5	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	oil	mp5	10	TJ+	0.951 [0.945, 0.958]	0.667	False
corn	oil	mp5	10	TJ+N	0.948 [0.941,	0.637	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					0.954]		
corn	oil	mp5	10	Transport	0.884 [0.872, 0.895]	0.651	True
corn	oil	mp5	10	WCP-S	0.590 [0.573, 0.605]	∞	True
corn	oil	mp5	15	ASTM-PI	0.211 [0.200, 0.222]	1.22	True
corn	oil	mp5	15	Gauss-LOO	0.906 [0.898, 0.914]	0.482	False
corn	oil	mp5	15	SCP-S	0.500 [0.486, 0.514]	0.777	True
corn	oil	mp5	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	oil	mp5	15	TJ+	0.957 [0.950, 0.963]	0.556	False
corn	oil	mp5	15	TJ+N	0.951 [0.944, 0.958]	0.523	False
corn	oil	mp5	15	Transport	0.876 [0.864, 0.889]	0.556	True
corn	oil	mp5	15	WCP-S	0.581 [0.565, 0.597]	2.92	True
corn	oil	mp5	20	ASTM-PI	0.208 [0.196, 0.221]	1.22	True
corn	oil	mp5	20	Gauss-LOO	0.903 [0.894, 0.912]	0.476	False
corn	oil	mp5	20	SCP-S	0.519 [0.503, 0.535]	0.745	True
corn	oil	mp5	20	SCP-T-split-N	0.915 [0.905, 0.925]	0.53	False
corn	oil	mp5	20	TJ+	0.917 [0.909, 0.926]	0.47	False
corn	oil	mp5	20	TJ+N	0.921 [0.913, 0.929]	0.447	False
corn	oil	mp5	20	Transport	0.881 [0.869, 0.894]	0.54	True
corn	oil	mp5	20	WCP-S	0.593 [0.577, 0.609]	0.911	True
corn	oil	mp6	10	ASTM-PI	0.201 [0.190, 0.211]	1.27	True
corn	oil	mp6	10	Gauss-LOO	0.901 [0.890, 0.911]	0.596	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
corn	oil	mp6	10	SCP-S	0.473 [0.458, 0.488]	0.832	True
corn	oil	mp6	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	oil	mp6	10	TJ+	0.946 [0.939, 0.953]	0.642	False
corn	oil	mp6	10	TJ+N	0.947 [0.940, 0.953]	0.612	False
corn	oil	mp6	10	Transport	0.921 [0.912, 0.930]	0.679	False
corn	oil	mp6	10	WCP-S	0.610 [0.595, 0.626]	∞	True
corn	oil	mp6	15	ASTM-PI	0.215 [0.203, 0.226]	1.18	True
corn	oil	mp6	15	Gauss-LOO	0.913 [0.904, 0.921]	0.49	False
corn	oil	mp6	15	SCP-S	0.517 [0.501, 0.533]	0.724	True
corn	oil	mp6	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	oil	mp6	15	TJ+	0.961 [0.956, 0.967]	0.574	False
corn	oil	mp6	15	TJ+N	0.959 [0.954, 0.965]	0.551	False
corn	oil	mp6	15	Transport	0.920 [0.910, 0.930]	0.582	False
corn	oil	mp6	15	WCP-S	0.600 [0.583, 0.616]	3.27	True
corn	oil	mp6	20	ASTM-PI	0.229 [0.216, 0.243]	1.15	True
corn	oil	mp6	20	Gauss-LOO	0.906 [0.897, 0.915]	0.461	False
corn	oil	mp6	20	SCP-S	0.538 [0.522, 0.554]	0.711	True
corn	oil	mp6	20	SCP-T-split-N	0.914 [0.904, 0.924]	0.51	False
corn	oil	mp6	20	TJ+	0.924 [0.916, 0.932]	0.468	False
corn	oil	mp6	20	TJ+N	0.927 [0.919, 0.934]	0.444	False
corn	oil	mp6	20	Transport	0.924	0.569	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					[0.914, 0.934]		
corn	oil	mp6	20	WCP-S	0.614 [0.597, 0.632]	1.08	True
corn	protein	mp5	10	ASTM-PI	0.381 [0.363, 0.399]	1.77	True
corn	protein	mp5	10	Gauss-LOO	0.906 [0.897, 0.915]	0.92	False
corn	protein	mp5	10	SCP-S	0.618 [0.599, 0.636]	1.07	True
corn	protein	mp5	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	protein	mp5	10	TJ+	0.933 [0.925, 0.940]	0.991	False
corn	protein	mp5	10	TJ+N	0.926 [0.919, 0.934]	1.02	False
corn	protein	mp5	10	Transport	0.883 [0.872, 0.894]	0.964	True
corn	protein	mp5	10	WCP-S	0.709 [0.693, 0.725]	∞	True
corn	protein	mp5	15	ASTM-PI	0.413 [0.395, 0.432]	1.52	True
corn	protein	mp5	15	Gauss-LOO	0.909 [0.900, 0.917]	0.811	False
corn	protein	mp5	15	SCP-S	0.667 [0.651, 0.684]	0.93	True
corn	protein	mp5	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	protein	mp5	15	TJ+	0.951 [0.945, 0.958]	0.927	False
corn	protein	mp5	15	TJ+N	0.944 [0.938, 0.951]	0.943	False
corn	protein	mp5	15	Transport	0.897 [0.886, 0.908]	0.839	False
corn	protein	mp5	15	WCP-S	0.727 [0.709, 0.742]	1.71	True
corn	protein	mp5	20	ASTM-PI	0.419 [0.399, 0.440]	1.48	True
corn	protein	mp5	20	Gauss-LOO	0.909 [0.900, 0.919]	0.745	False
corn	protein	mp5	20	SCP-S	0.670 [0.651,	0.875	True

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					0.689]		
corn	protein	mp5	20	SCP-T-split-N	0.910 [0.899, 0.920]	0.945	False
corn	protein	mp5	20	TJ+	0.915 [0.906, 0.925]	0.774	False
corn	protein	mp5	20	TJ+N	0.917 [0.908, 0.926]	0.793	False
corn	protein	mp5	20	Transport	0.898 [0.886, 0.910]	0.811	False
corn	protein	mp5	20	WCP-S	0.720 [0.702, 0.738]	1.28	True
corn	protein	mp6	10	ASTM-PI	0.393 [0.374, 0.411]	1.66	True
corn	protein	mp6	10	Gauss-LOO	0.910 [0.901, 0.919]	0.886	False
corn	protein	mp6	10	SCP-S	0.626 [0.608, 0.644]	1.03	True
corn	protein	mp6	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	protein	mp6	10	TJ+	0.942 [0.935, 0.948]	0.968	False
corn	protein	mp6	10	TJ+N	0.933 [0.926, 0.939]	0.999	False
corn	protein	mp6	10	Transport	0.913 [0.903, 0.924]	0.964	False
corn	protein	mp6	10	WCP-S	0.730 [0.714, 0.746]	∞	True
corn	protein	mp6	15	ASTM-PI	0.427 [0.408, 0.446]	1.46	True
corn	protein	mp6	15	Gauss-LOO	0.907 [0.899, 0.915]	0.805	False
corn	protein	mp6	15	SCP-S	0.670 [0.653, 0.686]	0.936	True
corn	protein	mp6	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	protein	mp6	15	TJ+	0.955 [0.949, 0.960]	0.904	False
corn	protein	mp6	15	TJ+N	0.950 [0.943, 0.956]	0.934	False
corn	protein	mp6	15	Transport	0.913 [0.902, 0.923]	0.873	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
corn	protein	mp6	15	WCP-S	0.733 [0.717, 0.750]	1.77	True
corn	protein	mp6	20	ASTM-PI	0.421 [0.401, 0.442]	1.42	True
corn	protein	mp6	20	Gauss-LOO	0.906 [0.897, 0.916]	0.746	False
corn	protein	mp6	20	SCP-S	0.682 [0.664, 0.700]	0.894	True
corn	protein	mp6	20	SCP-T-split-N	0.901 [0.890, 0.911]	0.896	False
corn	protein	mp6	20	TJ+	0.920 [0.911, 0.929]	0.775	False
corn	protein	mp6	20	TJ+N	0.920 [0.911, 0.929]	0.783	False
corn	protein	mp6	20	Transport	0.908 [0.897, 0.919]	0.849	False
corn	protein	mp6	20	WCP-S	0.738 [0.720, 0.755]	1.21	True
corn	starch	mp5	10	ASTM-PI	0.273 [0.259, 0.288]	4.88	True
corn	starch	mp5	10	Gauss-LOO	0.900 [0.890, 0.910]	2.37	False
corn	starch	mp5	10	SCP-S	0.567 [0.551, 0.584]	2.98	True
corn	starch	mp5	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	starch	mp5	10	TJ+	0.938 [0.930, 0.945]	2.64	False
corn	starch	mp5	10	TJ+N	0.932 [0.925, 0.940]	2.45	False
corn	starch	mp5	10	Transport	0.871 [0.860, 0.882]	2.55	True
corn	starch	mp5	10	WCP-S	0.708 [0.692, 0.724]	∞	True
corn	starch	mp5	15	ASTM-PI	0.282 [0.266, 0.297]	4.61	True
corn	starch	mp5	15	Gauss-LOO	0.904 [0.895, 0.913]	2.12	False
corn	starch	mp5	15	SCP-S	0.603 [0.587, 0.619]	2.79	True
corn	starch	mp5	15	SCP-T-	1.000	∞	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
				split-N	[1.000, 1.000]		
corn	starch	mp5	15	TJ+	0.946 [0.939, 0.953]	2.53	False
corn	starch	mp5	15	TJ+N	0.948 [0.941, 0.954]	2.37	False
corn	starch	mp5	15	Transport	0.863 [0.850, 0.875]	2.25	True
corn	starch	mp5	15	WCP-S	0.705 [0.690, 0.720]	∞	True
corn	starch	mp5	20	ASTM-PI	0.290 [0.274, 0.306]	4.44	True
corn	starch	mp5	20	Gauss-LOO	0.907 [0.898, 0.917]	1.98	False
corn	starch	mp5	20	SCP-S	0.619 [0.601, 0.636]	2.55	True
corn	starch	mp5	20	SCP-T-split-N	0.908 [0.897, 0.918]	2.28	False
corn	starch	mp5	20	TJ+	0.910 [0.901, 0.919]	2.03	False
corn	starch	mp5	20	TJ+N	0.909 [0.899, 0.919]	1.95	False
corn	starch	mp5	20	Transport	0.862 [0.850, 0.874]	2.06	True
corn	starch	mp5	20	WCP-S	0.699 [0.683, 0.716]	5.04	True
corn	starch	mp6	10	ASTM-PI	0.269 [0.255, 0.283]	5.01	True
corn	starch	mp6	10	Gauss-LOO	0.894 [0.883, 0.904]	2.41	False
corn	starch	mp6	10	SCP-S	0.559 [0.541, 0.576]	3.19	True
corn	starch	mp6	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	starch	mp6	10	TJ+	0.928 [0.919, 0.936]	2.6	False
corn	starch	mp6	10	TJ+N	0.926 [0.918, 0.935]	2.47	False
corn	starch	mp6	10	Transport	0.893 [0.883, 0.903]	2.48	False
corn	starch	mp6	10	WCP-S	0.700 [0.685,	∞	True

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					0.716]		
corn	starch	mp6	15	ASTM-PI	0.292 [0.276, 0.307]	4.63	True
corn	starch	mp6	15	Gauss-LOO	0.902 [0.894, 0.910]	2.14	False
corn	starch	mp6	15	SCP-S	0.606 [0.590, 0.622]	2.79	True
corn	starch	mp6	15	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
corn	starch	mp6	15	TJ+	0.953 [0.947, 0.958]	2.52	False
corn	starch	mp6	15	TJ+N	0.951 [0.944, 0.957]	2.41	False
corn	starch	mp6	15	Transport	0.893 [0.883, 0.903]	2.31	False
corn	starch	mp6	15	WCP-S	0.700 [0.684, 0.715]	∞	True
corn	starch	mp6	20	ASTM-PI	0.294 [0.278, 0.310]	4.38	True
corn	starch	mp6	20	Gauss-LOO	0.904 [0.894, 0.914]	1.95	False
corn	starch	mp6	20	SCP-S	0.616 [0.600, 0.633]	2.55	True
corn	starch	mp6	20	SCP-T-split-N	0.904 [0.894, 0.914]	2.25	False
corn	starch	mp6	20	TJ+	0.908 [0.898, 0.917]	2.02	False
corn	starch	mp6	20	TJ+N	0.908 [0.898, 0.917]	1.93	False
corn	starch	mp6	20	Transport	0.881 [0.868, 0.893]	2.09	True
corn	starch	mp6	20	WCP-S	0.696 [0.680, 0.712]	5.21	True
mango	dry_matter	S4	10	ASTM-PI	0.726 [0.720, 0.731]	5.27	True
mango	dry_matter	S4	10	Gauss-LOO	0.904 [0.892, 0.914]	5.02	False
mango	dry_matter	S4	10	SCP-S	0.707 [0.701, 0.712]	5.43	True
mango	dry_matter	S4	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
mango	dry_matter	S4	10	TJ+	0.910 [0.899, 0.921]	5.15	False
mango	dry_matter	S4	10	TJ+N	0.910 [0.899, 0.921]	5.06	False
mango	dry_matter	S4	10	WCP-S	0.715 [0.710, 0.720]	∞	True
mango	dry_matter	S4	20	ASTM-PI	0.749 [0.746, 0.751]	4.94	True
mango	dry_matter	S4	20	Gauss-LOO	0.903 [0.895, 0.911]	4.5	False
mango	dry_matter	S4	20	SCP-S	0.730 [0.727, 0.733]	5.09	True
mango	dry_matter	S4	20	SCP-T-split-N	0.905 [0.894, 0.915]	4.89	False
mango	dry_matter	S4	20	TJ+	0.903 [0.894, 0.910]	4.54	False
mango	dry_matter	S4	20	TJ+N	0.902 [0.893, 0.909]	4.46	False
mango	dry_matter	S4	20	WCP-S	0.738 [0.735, 0.740]	∞	True
mango	dry_matter	S4	30	ASTM-PI	0.755 [0.753, 0.757]	4.84	True
mango	dry_matter	S4	30	Gauss-LOO	0.901 [0.895, 0.907]	4.39	False
mango	dry_matter	S4	30	SCP-S	0.736 [0.733, 0.738]	4.98	True
mango	dry_matter	S4	30	SCP-T-split-N	0.933 [0.925, 0.940]	4.73	False
mango	dry_matter	S4	30	TJ+	0.900 [0.894, 0.906]	4.4	False
mango	dry_matter	S4	30	TJ+N	0.900 [0.894, 0.906]	4.34	False
mango	dry_matter	S4	30	WCP-S	0.743 [0.741, 0.745]	∞	True
mango	dry_matter	S4	50	ASTM-PI	0.762 [0.761, 0.764]	4.7	True
mango	dry_matter	S4	50	Gauss-LOO	0.908 [0.904, 0.913]	4.26	False
mango	dry_matter	S4	50	SCP-S	0.743 [0.742, 0.744]	4.84	True
mango	dry_matter	S4	50	SCP-T-	0.929	4.38	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
				split-N	[0.923, 0.934]		
mango	dry_matter	S4	50	TJ+	0.903 [0.899, 0.908]	4.26	False
mango	dry_matter	S4	50	TJ+N	0.904 [0.899, 0.908]	4.2	False
mango	dry_matter	S4	50	WCP-S	0.750 [0.749, 0.751]	∞	True
tablets	assay	2	10	ASTM-PI	0.846 [0.840, 0.852]	27.3	True
tablets	assay	2	10	Gauss-LOO	0.912 [0.904, 0.919]	31.2	False
tablets	assay	2	10	SCP-S	0.814 [0.807, 0.819]	28.8	True
tablets	assay	2	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
tablets	assay	2	10	TJ+	0.945 [0.940, 0.951]	33.1	False
tablets	assay	2	10	TJ+N	0.946 [0.940, 0.950]	31.2	False
tablets	assay	2	10	Transport	0.975 [0.973, 0.977]	35.8	False
tablets	assay	2	10	WCP-S	0.853 [0.846, 0.859]	∞	True
tablets	assay	2	20	ASTM-PI	0.927 [0.924, 0.930]	21.6	False
tablets	assay	2	20	Gauss-LOO	0.915 [0.910, 0.920]	23.1	False
tablets	assay	2	20	SCP-S	0.909 [0.907, 0.912]	21.7	False
tablets	assay	2	20	SCP-T-split-N	0.914 [0.906, 0.920]	28.2	False
tablets	assay	2	20	TJ+	0.917 [0.912, 0.922]	23.3	False
tablets	assay	2	20	TJ+N	0.918 [0.913, 0.923]	21.9	False
tablets	assay	2	20	Transport	0.950 [0.947, 0.952]	23.8	False
tablets	assay	2	20	WCP-S	0.926 [0.923, 0.929]	23.5	False
tablets	assay	2	30	ASTM-PI	0.941 [0.940,	21.1	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					0.943]		
tablets	assay	2	30	Gauss-LOO	0.914 [0.909, 0.919]	22.1	False
tablets	assay	2	30	SCP-S	0.926 [0.924, 0.928]	21	False
tablets	assay	2	30	SCP-T-split-N	0.933 [0.927, 0.939]	24.1	False
tablets	assay	2	30	TJ+	0.905 [0.900, 0.910]	21.7	False
tablets	assay	2	30	TJ+N	0.904 [0.899, 0.909]	20.4	False
tablets	assay	2	30	Transport	0.947 [0.945, 0.949]	22.2	False
tablets	assay	2	30	WCP-S	0.939 [0.937, 0.941]	21.8	False
tablets	assay	2	50	ASTM-PI	0.952 [0.951, 0.953]	20.8	False
tablets	assay	2	50	Gauss-LOO	0.926 [0.922, 0.929]	21.1	False
tablets	assay	2	50	SCP-S	0.938 [0.936, 0.939]	20.5	False
tablets	assay	2	50	SCP-T-split-N	0.922 [0.917, 0.927]	21.1	False
tablets	assay	2	50	TJ+	0.906 [0.902, 0.910]	20.5	False
tablets	assay	2	50	TJ+N	0.905 [0.901, 0.909]	19.4	False
tablets	assay	2	50	Transport	0.950 [0.949, 0.952]	21.3	False
tablets	assay	2	50	WCP-S	0.949 [0.947, 0.950]	21.1	False
tablets	hardness	2	10	ASTM-PI	0.949 [0.949, 0.950]	4.75	False
tablets	hardness	2	10	Gauss-LOO	0.895 [0.888, 0.902]	4.75	False
tablets	hardness	2	10	SCP-S	0.938 [0.937, 0.939]	4.66	False
tablets	hardness	2	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
tablets	hardness	2	10	TJ+	0.910 [0.903, 0.917]	4.86	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
tablets	hardness	2	10	TJ+N	0.911 [0.904, 0.918]	4.46	False
tablets	hardness	2	10	Transport	0.938 [0.938, 0.939]	4.67	False
tablets	hardness	2	10	WCP-S	0.938 [0.938, 0.939]	4.66	False
tablets	hardness	2	20	ASTM-PI	0.949 [0.949, 0.949]	4.77	False
tablets	hardness	2	20	Gauss-LOO	0.906 [0.901, 0.911]	4.66	False
tablets	hardness	2	20	SCP-S	0.939 [0.938, 0.939]	4.68	False
tablets	hardness	2	20	SCP-T-split-N	0.910 [0.902, 0.918]	4.49	False
tablets	hardness	2	20	TJ+	0.910 [0.904, 0.915]	4.74	False
tablets	hardness	2	20	TJ+N	0.910 [0.904, 0.916]	4.35	False
tablets	hardness	2	20	Transport	0.939 [0.939, 0.939]	4.69	False
tablets	hardness	2	20	WCP-S	0.939 [0.938, 0.939]	4.68	False
tablets	hardness	2	30	ASTM-PI	0.949 [0.948, 0.949]	4.77	False
tablets	hardness	2	30	Gauss-LOO	0.906 [0.902, 0.910]	4.64	False
tablets	hardness	2	30	SCP-S	0.939 [0.938, 0.939]	4.68	False
tablets	hardness	2	30	SCP-T-split-N	0.934 [0.928, 0.940]	4.54	False
tablets	hardness	2	30	TJ+	0.904 [0.899, 0.909]	4.68	False
tablets	hardness	2	30	TJ+N	0.904 [0.899, 0.908]	4.3	False
tablets	hardness	2	30	Transport	0.939 [0.939, 0.939]	4.7	False
tablets	hardness	2	30	WCP-S	0.939 [0.938, 0.939]	4.68	False
tablets	hardness	2	50	ASTM-PI	0.949 [0.948, 0.949]	4.78	False
tablets	hardness	2	50	Gauss-	0.907	4.61	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
				LOO	[0.904, 0.911]		
tablets	hardness	2	50	SCP-S	0.939 [0.938, 0.939]	4.69	False
tablets	hardness	2	50	SCP-T-split-N	0.919 [0.915, 0.924]	4.38	False
tablets	hardness	2	50	TJ+	0.901 [0.897, 0.905]	4.64	False
tablets	hardness	2	50	TJ+N	0.900 [0.896, 0.903]	4.28	False
tablets	hardness	2	50	Transport	0.939 [0.939, 0.939]	4.7	False
tablets	hardness	2	50	WCP-S	0.939 [0.938, 0.939]	4.69	False
tablets	weight	2	10	ASTM-PI	0.911 [0.911, 0.912]	17	False
tablets	weight	2	10	Gauss-LOO	0.907 [0.899, 0.914]	18	False
tablets	weight	2	10	SCP-S	0.895 [0.894, 0.896]	17	True
tablets	weight	2	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	False
tablets	weight	2	10	TJ+	0.914 [0.906, 0.921]	18.4	False
tablets	weight	2	10	TJ+N	0.919 [0.913, 0.926]	18.8	False
tablets	weight	2	10	Transport	0.887 [0.886, 0.888]	17.1	True
tablets	weight	2	10	WCP-S	0.895 [0.894, 0.896]	17	True
tablets	weight	2	20	ASTM-PI	0.910 [0.910, 0.911]	17.1	False
tablets	weight	2	20	Gauss-LOO	0.905 [0.899, 0.910]	17.6	False
tablets	weight	2	20	SCP-S	0.895 [0.894, 0.895]	17.1	True
tablets	weight	2	20	SCP-T-split-N	0.914 [0.906, 0.921]	19	False
tablets	weight	2	20	TJ+	0.910 [0.904, 0.915]	17.6	False
tablets	weight	2	20	TJ+N	0.906 [0.901,	18.3	False

Data	Response	Target	m	Method	Coverage [95% CI]	Median IS	Under-covering
					0.912]		
tablets	weight	2	20	Transport	0.885 [0.884, 0.886]	17.2	True
tablets	weight	2	20	WCP-S	0.895 [0.894, 0.895]	17.1	True
tablets	weight	2	30	ASTM-PI	0.909 [0.909, 0.910]	17.2	False
tablets	weight	2	30	Gauss-LOO	0.896 [0.892, 0.901]	17.5	False
tablets	weight	2	30	SCP-S	0.894 [0.894, 0.895]	17.2	True
tablets	weight	2	30	SCP-T-split-N	0.932 [0.927, 0.938]	19.4	False
tablets	weight	2	30	TJ+	0.900 [0.895, 0.905]	17.5	False
tablets	weight	2	30	TJ+N	0.898 [0.893, 0.903]	18.1	False
tablets	weight	2	30	Transport	0.884 [0.883, 0.885]	17.2	True
tablets	weight	2	30	WCP-S	0.894 [0.894, 0.895]	17.2	True
tablets	weight	2	50	ASTM-PI	0.909 [0.909, 0.910]	17.1	False
tablets	weight	2	50	Gauss-LOO	0.907 [0.904, 0.911]	17.4	False
tablets	weight	2	50	SCP-S	0.895 [0.894, 0.895]	17.2	True
tablets	weight	2	50	SCP-T-split-N	0.926 [0.922, 0.930]	18.5	False
tablets	weight	2	50	TJ+	0.906 [0.902, 0.909]	17.4	False
tablets	weight	2	50	TJ+N	0.905 [0.901, 0.908]	18.1	False
tablets	weight	2	50	Transport	0.883 [0.883, 0.884]	17.2	True
tablets	weight	2	50	WCP-S	0.895 [0.894, 0.895]	17.2	True

Table S2. All transfer pipelines

Table S2. Coverage (mean and minimum over scenarios), median interval score, RMSEP and under-covering scenarios for every transfer pipeline (α = 0.10).

dataset	kind	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
corn	ds	ASTM-PI	0.131	0.035	3.884	0.392	24	24
corn	ds	Gauss-LOO	0.897	0.884	1.354	0.392	4	24
corn	ds	SCP-S	0.294	0.066	3.534	0.392	24	24
corn	ds	SCP-T-split-N	0.969	0.896	inf	0.392	0	24
corn	ds	TJ+	0.932	0.91	1.435	0.392	0	24
corn	ds	TJ+N	0.93	0.907	1.472	0.392	0	24
corn	none	ASTM-PI	0.024	0.0	21.598	1.144	24	24
corn	none	Gauss-LOO	0.992	0.98	4.181	1.144	0	24
corn	none	SCP-S	0.041	0.0	21.556	1.144	24	24
corn	none	SCP-T-split-N	0.969	0.897	inf	1.144	0	24
corn	none	TJ+	0.916	0.898	3.127	1.144	0	24
corn	none	TJ+N	0.916	0.899	3.149	1.144	0	24
corn	pds	ASTM-PI	0.233	0.026	3.045	0.292	24	24
corn	pds	Gauss-LOO	0.904	0.893	1.055	0.292	0	24
corn	pds	SCP-S	0.453	0.051	1.809	0.292	24	24
corn	pds	SCP-T-split-N	0.97	0.901	inf	0.292	0	24
corn	pds	TJ+	0.933	0.908	1.107	0.292	0	24
corn	pds	TJ+N	0.931	0.908	1.138	0.292	0	24
corn	sbc	ASTM-PI	0.409	0.136	1.743	0.221	24	24
corn	sbc	Gauss-LOO	0.9	0.887	0.849	0.221	2	24
corn	sbc	SCP-S	0.515	0.073	1.424	0.221	24	24
corn	sbc	SCP-T-split-N	0.971	0.907	inf	0.221	0	24
corn	sbc	TJ+	0.917	0.899	0.868	0.221	0	24
corn	sbc	TJ+N	0.918	0.902	0.863	0.221	0	24
mango	none	ASTM-PI	0.753	0.753	4.762	1.052	4	4
mango	none	Gauss-LOO	0.905	0.901	4.282	1.052	0	4
mango	none	SCP-S	0.735	0.735	4.915	1.052	4	4
mango	none	SCP-T-split-N	0.941	0.903	4.333	1.052	0	4
mango	none	TJ+	0.904	0.897	4.29	1.052	0	4
mango	none	TJ+N	0.905	0.901	4.176	1.052	0	4
mango	sbc	ASTM-PI	0.748	0.726	4.888	1.096	4	4
mango	sbc	Gauss-LOO	0.904	0.902	4.446	1.096	0	4
mango	sbc	SCP-S	0.729	0.707	5.034	1.096	4	4
mango	sbc	SCP-T-split-N	0.942	0.905	4.807	1.096	0	4
mango	sbc	TJ+	0.904	0.9	4.472	1.096	0	4
mango	sbc	TJ+N	0.904	0.9	4.398	1.096	0	4
tablets	ds	ASTM-PI	0.897	0.658	16.999	3.935	3	12
tablets	ds	Gauss-LOO	0.908	0.897	17.387	3.935	0	12
tablets	ds	SCP-S	0.88	0.627	16.981	3.935	7	12
tablets	ds	SCP-T-split-N	0.942	0.908	22.316	3.935	0	12
tablets	ds	TJ+	0.91	0.9	17.416	3.935	0	12

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dataset	kind	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
tablets	ds	TJ+N	0.909	0.898	18.232	3.935	0	12
tablets	none	ASTM-PI	0.905	0.865	18.313	4.285	8	12
tablets	none	Gauss-LOO	0.902	0.875	18.554	4.285	4	12
tablets	none	SCP-S	0.898	0.863	18.366	4.285	8	12
tablets	none	SCP-T-split-N	0.943	0.909	29.065	4.285	0	12
tablets	none	TJ+	0.906	0.9	18.644	4.285	0	12
tablets	none	TJ+N	0.906	0.9	18.999	4.285	0	12
tablets	pds	ASTM-PI	0.925	0.846	17.138	3.492	1	12
tablets	pds	Gauss-LOO	0.908	0.895	17.508	3.492	0	12
tablets	pds	SCP-S	0.91	0.814	17.152	3.492	5	12
tablets	pds	SCP-T-split-N	0.942	0.91	20.254	3.492	0	12
tablets	pds	TJ+	0.911	0.9	17.532	3.492	0	12
tablets	pds	TJ+N	0.91	0.898	18.224	3.492	0	12
tablets	sbc	ASTM-PI	0.904	0.798	15.911	4.084	4	12
tablets	sbc	Gauss-LOO	0.899	0.879	16.176	4.084	2	12
tablets	sbc	SCP-S	0.891	0.78	15.747	4.084	5	12
tablets	sbc	SCP-T-split-N	0.943	0.907	25.828	4.084	0	12
tablets	sbc	TJ+	0.906	0.899	16.264	4.084	0	12
tablets	sbc	TJ+N	0.906	0.9	16.462	4.084	0	12

Table S3. Sensitivity to transfer and model settings

Table S3. Sensitivity to the PDS half-window (hw1 = 3 channels, hw3 = 7, hw5 = 11; the primary analysis used 5 channels) and to the number of latent variables (lv ±2).

dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
corn	hw1	ASTM-PI	0.25	0.033	2.655	0.237	4	4
corn	hw1	Gauss-LOO	0.903	0.89	0.932	0.237	0	4
corn	hw1	SCP-S	0.496	0.064	1.566	0.237	4	4
corn	hw1	SCP-T-split-N	1.0	1.0	inf	0.237	0	4
corn	hw1	TJ+	0.941	0.93	1.057	0.237	0	4
corn	hw1	TJ+N	0.941	0.927	1.083	0.237	0	4
corn	hw1	WCP-S	0.588	0.217	inf	0.237	4	4
corn	hw3	ASTM-PI	0.213	0.028	3.314	0.348	4	4
corn	hw3	Gauss-LOO	0.909	0.896	1.21	0.348	0	4
corn	hw3	SCP-S	0.404	0.056	2.363	0.348	4	4
corn	hw3	SCP-T-split-N	1.0	1.0	inf	0.348	0	4
corn	hw3	TJ+	0.962	0.951	1.313	0.348	0	4
corn	hw3	TJ+N	0.954	0.938	1.272	0.348	0	4
corn	hw3	WCP-S	0.509	0.217	inf	0.348	4	4
corn	hw5	ASTM-PI	0.17	0.032	4.362	0.503	4	4
corn	hw5	Gauss-LOO	0.903	0.896	1.549	0.503	0	4
corn	hw5	SCP-S	0.314	0.053	4.002	0.503	4	4
corn	hw5	SCP-T-split-N	1.0	1.0	inf	0.503	0	4
corn	hw5	TJ+	0.967	0.955	1.735	0.503	0	4
corn	hw5	TJ+N	0.962	0.95	1.512	0.503	0	4
corn	hw5	WCP-S	0.44	0.233	inf	0.503	4	4

dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
corn	lv+2	ASTM-PI	0.189	0.033	3.136	0.283	4	4
corn	lv+2	Gauss-LOO	0.905	0.893	1.047	0.283	0	4
corn	lv+2	SCP-S	0.436	0.059	1.872	0.283	4	4
corn	lv+2	SCP-T-split-N	1.0	1.0	inf	0.283	0	4
corn	lv+2	TJ+	0.952	0.943	1.18	0.283	0	4
corn	lv+2	TJ+N	0.949	0.939	1.2	0.283	0	4
corn	lv+2	WCP-S	0.552	0.237	inf	0.283	4	4
corn	lv-2	ASTM-PI	0.313	0.051	2.488	0.275	4	4
corn	lv-2	Gauss-LOO	0.904	0.894	1.06	0.275	0	4
corn	lv-2	SCP-S	0.516	0.083	1.616	0.275	4	4
corn	lv-2	SCP-T-split-N	1.0	1.0	inf	0.275	0	4
corn	lv-2	TJ+	0.951	0.943	1.161	0.275	0	4
corn	lv-2	TJ+N	0.944	0.933	1.191	0.275	0	4
corn	lv-2	WCP-S	0.589	0.203	2.741	0.275	4	4
mango	lv+2	ASTM-PI	0.737	0.737	5.054	1.108	1	1
mango	lv+2	Gauss-LOO	0.898	0.898	4.538	1.108	0	1
mango	lv+2	SCP-S	0.721	0.721	5.186	1.108	1	1
mango	lv+2	SCP-T-split-N	0.902	0.902	4.989	1.108	0	1
mango	lv+2	TJ+	0.899	0.899	4.612	1.108	0	1
mango	lv+2	TJ+N	0.898	0.898	4.535	1.108	0	1
mango	lv+2	WCP-S	0.734	0.734	inf	1.108	1	1
mango	lv-2	ASTM-PI	0.757	0.757	5.016	1.106	1	1
mango	lv-2	Gauss-LOO	0.896	0.896	4.587	1.106	0	1
mango	lv-2	SCP-S	0.745	0.745	5.117	1.106	1	1
mango	lv-2	SCP-T-split-N	0.906	0.906	4.968	1.106	0	1
mango	lv-2	TJ+	0.9	0.9	4.642	1.106	0	1
mango	lv-2	TJ+N	0.899	0.899	4.544	1.106	0	1
mango	lv-2	WCP-S	0.793	0.793	inf	1.106	1	1
tablets	hw1	ASTM-PI	0.943	0.943	20.805	4.767	0	1
tablets	hw1	Gauss-LOO	0.911	0.911	21.754	4.767	0	1
tablets	hw1	SCP-S	0.928	0.928	20.589	4.767	0	1
tablets	hw1	SCP-T-split-N	0.912	0.912	23.102	4.767	0	1
tablets	hw1	TJ+	0.913	0.913	21.903	4.767	0	1
tablets	hw1	TJ+N	0.916	0.916	20.449	4.767	0	1
tablets	hw1	WCP-S	0.939	0.939	22.007	4.767	0	1
tablets	hw3	ASTM-PI	0.906	0.906	22.434	5.39	0	1
tablets	hw3	Gauss-LOO	0.91	0.91	24.192	5.39	0	1
tablets	hw3	SCP-S	0.884	0.884	22.73	5.39	1	1
tablets	hw3	SCP-T-split-N	0.92	0.92	33.539	5.39	0	1
tablets	hw3	TJ+	0.916	0.916	24.22	5.39	0	1
tablets	hw3	TJ+N	0.916	0.916	23.243	5.39	0	1
tablets	hw3	WCP-S	0.909	0.909	24.208	5.39	0	1
tablets	hw5	ASTM-PI	0.867	0.867	25.273	5.98	1	1
tablets	hw5	Gauss-LOO	0.912	0.912	27.218	5.98	0	1
tablets	hw5	SCP-S	0.837	0.837	26.267	5.98	1	1
tablets	hw5	SCP-T-split-N	0.915	0.915	37.461	5.98	0	1
tablets	hw5	TJ+	0.925	0.925	27.284	5.98	0	1

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dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
tablets	hw5	TJ+N	0.924	0.924	26.505	5.98	0	1
tablets	hw5	WCP-S	0.87	0.87	29.995	5.98	1	1
tablets	lv+2	ASTM-PI	0.898	0.898	22.363	5.326	0	1
tablets	lv+2	Gauss-LOO	0.908	0.908	24.135	5.326	0	1
tablets	lv+2	SCP-S	0.876	0.876	23.175	5.326	1	1
tablets	lv+2	SCP-T-split-N	0.918	0.918	26.722	5.326	0	1
tablets	lv+2	TJ+	0.91	0.91	24.74	5.326	0	1
tablets	lv+2	TJ+N	0.918	0.918	21.987	5.326	0	1
tablets	lv+2	WCP-S	0.907	0.907	inf	5.326	0	1
tablets	lv-2	ASTM-PI	0.903	0.903	29.591	6.999	0	1
tablets	lv-2	Gauss-LOO	0.909	0.909	31.299	6.999	0	1
tablets	lv-2	SCP-S	0.912	0.912	29.915	6.999	0	1
tablets	lv-2	SCP-T-split-N	0.908	0.908	37.834	6.999	0	1
tablets	lv-2	TJ+	0.912	0.912	32.295	6.999	0	1
tablets	lv-2	TJ+N	0.916	0.916	28.684	6.999	0	1
tablets	lv-2	WCP-S	0.92	0.92	30.799	6.999	0	1

Table S4. Computational cost

Table S4. Wall-clock time and peak memory (single CPU core, median of 20 runs).

dataset	kind	m	n_test	n_channels	step	median_s	p90_s	peak_mem_MB	source_model_fit_s
corn	pds	20	10	700	point prediction only (1 transfer fit)	0.0046	0.0061	0.8785	0.1027
corn	pds	20	10	700	SCP-T-split (1 fit)	0.0047	0.0065	0.7389	0.1027
corn	pds	20	10	700	TJ+N incl. m LOO + 5 CV fits	0.1469	0.1572	1.2104	0.1027
tablets	pds	50	410	650	point prediction only (1 transfer fit)	0.0243	0.0298	14.3825	0.2743
tablets	pds	50	410	650	SCP-T-split (1 fit)	0.0259	0.0265	14.3831	0.2743
tablets	pds	50	410	650	TJ+N incl. m LOO + 5 CV fits	2.7857	2.9018	15.2063	0.2743
mango	sbc	50	675	103	point prediction only (1 transfer fit)	0.0004	0.0005	1.1246	1.4279
mango	sbc	50	675	103	SCP-T-split (1 fit)	0.0004	0.0004	1.1246	1.4279

dataset	kind	m	n_test	n_channels	step	median_s	p90_s	peak_mem_MB	source_model_fit_s
mango	sbc	50	675	103	TJ+N incl. m LOO + 5 CV fits	0.0635	0.0742	3.1352	1.4279

Table S5. Paired tests

Table S5. Paired Wilcoxon signed-rank tests of per-repetition interval score, TJ+N against each method: number of scenarios with a significantly lower (better) or higher (worse) score for TJ+N (Holm-adjusted $p < 0.05$ within scenario) and median matched-pairs rank-biserial correlation (negative = TJ+N better).

dataset	other	n_scen	n_sig_TJ_better	median_rank_biserial	n_sig_TJ_worse
corn	ASTM-PI	24	24	-0.978	0
corn	Gauss-LOO	24	4	0.143	14
corn	LF-conv	24	4	0.056	8
corn	SCP-S	24	24	-0.681	0
corn	SCP-T-split	24	8	-1.0	0
corn	SCP-T-split-N	24	8	-1.0	0
corn	TCV+N	24	24	-0.417	0
corn	TJ+	24	12	-0.157	1
corn	TJ+N[Q]	24	4	-0.335	0
corn	TJ+N[T2]	24	0	0.172	6
corn	TJ-insample-N	24	0	0.559	24
corn	TJ-naive-N	24	0	0.452	23
corn	Transport	24	20	-0.33	0
corn	WCP-S	24	21	-0.918	3
mango	ASTM-PI	4	4	-0.929	0
mango	Gauss-LOO	4	2	-0.26	1
mango	SCP-S	4	4	-0.965	0
mango	SCP-T-split	4	3	-0.861	0
mango	SCP-T-split-N	4	3	-0.771	0
mango	TCV+N	4	4	-0.186	0
mango	TJ+	4	4	-0.54	0
mango	TJ+N[Q]	4	4	-0.5	0
mango	TJ+N[T2]	4	1	-0.132	0
mango	TJ-insample-N	4	0	0.065	1
mango	TJ-naive-N	4	0	0.207	3
mango	WCP-S	4	0	-1.0	0
tablets	ASTM-PI	12	7	-0.18	5
tablets	Gauss-LOO	12	7	-0.55	4
tablets	LF-conv	12	7	-0.111	4
tablets	SCP-S	12	7	-0.144	5
tablets	SCP-T-split	12	6	-0.977	2
tablets	SCP-T-split-N	12	9	-0.799	0
tablets	TCV+N	12	4	0.038	3
tablets	TJ+	12	8	-0.816	4
tablets	TJ+N[Q]	12	8	-0.636	4
tablets	TJ+N[T2]	12	5	0.127	6
tablets	TJ-insample-N	12	7	-0.246	3
tablets	TJ-naive-N	12	2	0.172	7
tablets	Transport	12	8	-0.535	4
tablets	WCP-S	12	7	-0.624	5

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Table S6. Independent instrument shift (mango, E8)

Table S6. Mango instrument transfer (source instrument 15006; α = 0.10; 200 repetitions).

Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
pds	15014	10	ASTM-PI	0.295 [0.271, 0.321]	44.2	5.594	True
pds	15014	10	Gauss-LOO	0.928 [0.915, 0.940]	22	5.594	False
pds	15014	10	LF-conv	0.878 [0.861, 0.893]	23.6	5.594	True
pds	15014	10	SCP-S	0.280 [0.257, 0.304]	46	5.594	True
pds	15014	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	5.594	False
pds	15014	10	TJ+	0.972 [0.965, 0.979]	27.7	5.594	False
pds	15014	10	TJ+N	0.973 [0.968, 0.978]	23.6	5.594	False
pds	15014	10	Transport	0.974 [0.965, 0.983]	37.6	5.594	False
pds	15014	10	WCP-S	0.277 [0.254, 0.300]	∞	5.594	True
sbc	15014	10	ASTM-PI	0.738 [0.731, 0.743]	5.79	1.2	True
sbc	15014	10	Gauss-LOO	0.899 [0.886, 0.911]	5.37	1.2	False
sbc	15014	10	SCP-S	0.726 [0.720, 0.732]	5.9	1.2	True
sbc	15014	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	1.2	False
sbc	15014	10	TJ+	0.915 [0.904, 0.926]	5.6	1.2	False
sbc	15014	10	TJ+N	0.918 [0.907, 0.928]	5.55	1.2	False
sbc	15014	10	WCP-S	0.999 [0.999, 0.999]	∞	1.2	False
pds	15014	20	ASTM-PI	0.357 [0.334, 0.380]	29	4.017	True
pds	15014	20	Gauss-LOO	0.914 [0.903, 0.925]	14.8	4.017	False
pds	15014	20	LF-conv	0.894 [0.882,	15	4.017	False

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
				0.905]			
pds	15014	20	SCP-S	0.345 [0.322, 0.368]	30	4.017	True
pds	15014	20	SCP-T-split-N	0.913 [0.901, 0.924]	14.6	4.017	False
pds	15014	20	TJ+	0.933 [0.925, 0.940]	14.8	4.017	False
pds	15014	20	TJ+N	0.941 [0.934, 0.947]	11.9	4.017	False
pds	15014	20	Transport	0.958 [0.947, 0.968]	21.9	4.017	False
pds	15014	20	WCP-S	0.309 [0.286, 0.331]	52.2	4.017	True
sbc	15014	20	ASTM-PI	0.762 [0.759, 0.765]	5.4	1.141	True
sbc	15014	20	Gauss-LOO	0.901 [0.893, 0.909]	5.01	1.141	False
sbc	15014	20	SCP-S	0.751 [0.748, 0.754]	5.5	1.141	True
sbc	15014	20	SCP-T-split-N	0.912 [0.901, 0.923]	5.42	1.141	False
sbc	15014	20	TJ+	0.910 [0.900, 0.918]	5.13	1.141	False
sbc	15014	20	TJ+N	0.912 [0.903, 0.921]	5.08	1.141	False
sbc	15014	20	WCP-S	1.000 [0.999, 1.000]	∞	1.141	False
pds	15014	30	ASTM-PI	0.400 [0.373, 0.427]	23.8	3.421	True
pds	15014	30	Gauss-LOO	0.912 [0.903, 0.920]	12	3.421	False
pds	15014	30	LF-conv	0.888 [0.879, 0.898]	12.2	3.421	True
pds	15014	30	SCP-S	0.389 [0.362, 0.414]	24.4	3.421	True
pds	15014	30	SCP-T-split-N	0.939 [0.931, 0.946]	12.1	3.421	False
pds	15014	30	TJ+	0.922 [0.916, 0.928]	11.9	3.421	False
pds	15014	30	TJ+N	0.924 [0.918, 0.929]	10	3.421	False

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
pds	15014	30	Transport	0.955 [0.943, 0.965]	16	3.421	False
pds	15014	30	WCP-S	0.356 [0.331, 0.380]	35.3	3.421	True
sbc	15014	30	ASTM-PI	0.768 [0.765, 0.770]	5.33	1.127	True
sbc	15014	30	Gauss-LOO	0.903 [0.897, 0.910]	4.88	1.127	False
sbc	15014	30	SCP-S	0.756 [0.753, 0.759]	5.42	1.127	True
sbc	15014	30	SCP-T-split-N	0.938 [0.929, 0.945]	5.34	1.127	False
sbc	15014	30	TJ+	0.906 [0.899, 0.913]	4.93	1.127	False
sbc	15014	30	TJ+N	0.906 [0.898, 0.913]	4.91	1.127	False
sbc	15014	30	WCP-S	1.000 [0.999, 1.000]	∞	1.127	False
pds	15014	50	ASTM-PI	0.408 [0.382, 0.435]	18.9	3.43	True
pds	15014	50	Gauss-LOO	0.904 [0.897, 0.910]	9.91	3.43	False
pds	15014	50	LF-conv	0.885 [0.877, 0.893]	10.1	3.43	True
pds	15014	50	SCP-S	0.397 [0.373, 0.423]	19.4	3.43	True
pds	15014	50	SCP-T-split-N	0.920 [0.912, 0.927]	10.7	3.43	False
pds	15014	50	TJ+	0.907 [0.901, 0.912]	10.1	3.43	False
pds	15014	50	TJ+N	0.909 [0.905, 0.915]	8.42	3.43	False
pds	15014	50	Transport	0.945 [0.931, 0.958]	15.1	3.43	False
pds	15014	50	WCP-S	0.356 [0.332, 0.381]	21.8	3.43	True
sbc	15014	50	ASTM-PI	0.773 [0.772, 0.774]	5.28	1.115	True
sbc	15014	50	Gauss-LOO	0.896 [0.890, 0.901]	4.8	1.115	False
sbc	15014	50	SCP-S	0.762	5.38	1.115	True

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
				[0.760, 0.763]			
sbc	15014	50	SCP-T-split-N	0.924 [0.917, 0.931]	4.98	1.115	False
sbc	15014	50	TJ+	0.899 [0.893, 0.906]	4.84	1.115	False
sbc	15014	50	TJ+N	0.899 [0.893, 0.905]	4.77	1.115	False
sbc	15014	50	WCP-S	1.000 [1.000, 1.000]	∞	1.115	False
pds	15016	10	ASTM-PI	0.219 [0.204, 0.235]	66.2	7.401	True
pds	15016	10	Gauss-LOO	0.922 [0.907, 0.936]	29.5	7.401	False
pds	15016	10	LF-conv	0.859 [0.840, 0.877]	31.5	7.401	True
pds	15016	10	SCP-S	0.204 [0.188, 0.220]	68.9	7.401	True
pds	15016	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	7.401	False
pds	15016	10	TJ+	0.956 [0.945, 0.966]	36.5	7.401	False
pds	15016	10	TJ+N	0.965 [0.955, 0.972]	27.6	7.401	False
pds	15016	10	Transport	0.989 [0.984, 0.994]	69.2	7.401	False
pds	15016	10	WCP-S	0.151 [0.138, 0.165]	176	7.401	True
sbc	15016	10	ASTM-PI	0.767 [0.760, 0.772]	5.55	1.224	True
sbc	15016	10	Gauss-LOO	0.894 [0.881, 0.907]	5.33	1.224	False
sbc	15016	10	SCP-S	0.729 [0.723, 0.735]	5.88	1.224	True
sbc	15016	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	1.224	False
sbc	15016	10	TJ+	0.902 [0.887, 0.916]	5.64	1.224	False
sbc	15016	10	TJ+N	0.905 [0.890, 0.918]	6.34	1.224	False
sbc	15016	10	WCP-S	1.000 [1.000,	∞	1.224	False

Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
				1.000]			
pds	15016	20	ASTM-PI	0.262 [0.247, 0.277]	50	5.086	True
pds	15016	20	Gauss-LOO	0.921 [0.912, 0.929]	21.4	5.086	False
pds	15016	20	LF-conv	0.906 [0.896, 0.915]	21.5	5.086	False
pds	15016	20	SCP-S	0.248 [0.234, 0.263]	51.8	5.086	True
pds	15016	20	SCP-T-split-N	0.923 [0.912, 0.933]	22	5.086	False
pds	15016	20	TJ+	0.929 [0.921, 0.937]	21.8	5.086	False
pds	15016	20	TJ+N	0.942 [0.935, 0.948]	17.1	5.086	False
pds	15016	20	Transport	0.991 [0.986, 0.995]	44.1	5.086	False
pds	15016	20	WCP-S	0.159 [0.148, 0.170]	74.3	5.086	True
sbc	15016	20	ASTM-PI	0.790 [0.786, 0.793]	5.24	1.173	True
sbc	15016	20	Gauss-LOO	0.900 [0.890, 0.908]	5	1.173	False
sbc	15016	20	SCP-S	0.753 [0.750, 0.756]	5.49	1.173	True
sbc	15016	20	SCP-T-split-N	0.905 [0.894, 0.916]	6.09	1.173	False
sbc	15016	20	TJ+	0.905 [0.896, 0.914]	5.08	1.173	False
sbc	15016	20	TJ+N	0.905 [0.896, 0.914]	5.55	1.173	False
sbc	15016	20	WCP-S	1.000 [1.000, 1.000]	∞	1.173	False
pds	15016	30	ASTM-PI	0.293 [0.279, 0.308]	43.1	4.33	True
pds	15016	30	Gauss-LOO	0.916 [0.908, 0.923]	18	4.33	False
pds	15016	30	LF-conv	0.897 [0.889, 0.905]	18.5	4.33	False
pds	15016	30	SCP-S	0.280 [0.265, 0.294]	44.5	4.33	True

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
pds	15016	30	SCP-T-split-N	0.938 [0.930, 0.946]	17.9	4.33	False
pds	15016	30	TJ+	0.918 [0.911, 0.926]	18.1	4.33	False
pds	15016	30	TJ+N	0.915 [0.909, 0.922]	14.2	4.33	False
pds	15016	30	Transport	0.984 [0.977, 0.989]	32.7	4.33	False
pds	15016	30	WCP-S	0.168 [0.158, 0.178]	54.4	4.33	True
sbc	15016	30	ASTM-PI	0.797 [0.795, 0.800]	5.13	1.155	True
sbc	15016	30	Gauss-LOO	0.902 [0.895, 0.909]	4.89	1.155	False
sbc	15016	30	SCP-S	0.761 [0.758, 0.763]	5.36	1.155	True
sbc	15016	30	SCP-T-split-N	0.937 [0.928, 0.945]	6.08	1.155	False
sbc	15016	30	TJ+	0.900 [0.893, 0.908]	4.93	1.155	False
sbc	15016	30	TJ+N	0.901 [0.894, 0.909]	5.32	1.155	False
sbc	15016	30	WCP-S	1.000 [1.000, 1.000]	∞	1.155	False
pds	15016	50	ASTM-PI	0.317 [0.304, 0.331]	36.9	3.924	True
pds	15016	50	Gauss-LOO	0.911 [0.905, 0.917]	16.5	3.924	False
pds	15016	50	LF-conv	0.896 [0.890, 0.902]	16.4	3.924	False
pds	15016	50	SCP-S	0.304 [0.291, 0.318]	38	3.924	True
pds	15016	50	SCP-T-split-N	0.921 [0.913, 0.928]	14.6	3.924	False
pds	15016	50	TJ+	0.903 [0.898, 0.909]	16.2	3.924	False
pds	15016	50	TJ+N	0.907 [0.902, 0.913]	12.6	3.924	False
pds	15016	50	Transport	0.984 [0.980, 0.988]	26.4	3.924	False
pds	15016	50	WCP-S	0.175	46.1	3.924	True

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
				[0.167, 0.183]			
sbc	15016	50	ASTM-PI	0.803 [0.801, 0.805]	5.05	1.141	True
sbc	15016	50	Gauss-LOO	0.905 [0.901, 0.910]	4.79	1.141	False
sbc	15016	50	SCP-S	0.766 [0.763, 0.768]	5.28	1.141	True
sbc	15016	50	SCP-T-split-N	0.916 [0.908, 0.923]	5.41	1.141	False
sbc	15016	50	TJ+	0.902 [0.897, 0.908]	4.83	1.141	False
sbc	15016	50	TJ+N	0.903 [0.897, 0.908]	5.17	1.141	False
sbc	15016	50	WCP-S	1.000 [1.000, 1.000]	∞	1.141	False
pds	15029	10	ASTM-PI	0.332 [0.312, 0.352]	30.8	3.95	True
pds	15029	10	Gauss-LOO	0.923 [0.908, 0.938]	16.6	3.95	False
pds	15029	10	LF-conv	0.866 [0.849, 0.883]	16.9	3.95	True
pds	15029	10	SCP-S	0.320 [0.300, 0.340]	31.6	3.95	True
pds	15029	10	SCP-T-split-N	1.000 [1.000, 1.000]	∞	3.95	False
pds	15029	10	TJ+	0.961 [0.952, 0.968]	19.6	3.95	False
pds	15029	10	TJ+N	0.960 [0.953, 0.966]	17.6	3.95	False
pds	15029	10	Transport	0.992 [0.988, 0.996]	42.5	3.95	False
pds	15029	10	WCP-S	0.308 [0.286, 0.330]	∞	3.95	True
sbc	15029	10	ASTM-PI	0.773 [0.765, 0.780]	5.36	1.155	True
sbc	15029	10	Gauss-LOO	0.893 [0.880, 0.905]	5.31	1.155	False
sbc	15029	10	SCP-S	0.763 [0.757, 0.770]	5.42	1.155	True
sbc	15029	10	SCP-T-split-N	1.000 [1.000,	∞	1.155	False

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
				1.000]			
sbc	15029	10	TJ+	0.910 [0.897, 0.921]	5.51	1.155	False
sbc	15029	10	TJ+N	0.910 [0.897, 0.922]	5.66	1.155	False
sbc	15029	10	WCP-S	0.999 [0.999, 0.999]	∞	1.155	False
pds	15029	20	ASTM-PI	0.408 [0.388, 0.425]	20.4	2.779	True
pds	15029	20	Gauss-LOO	0.920 [0.910, 0.930]	10.4	2.779	False
pds	15029	20	LF-conv	0.900 [0.889, 0.910]	10.8	2.779	False
pds	15029	20	SCP-S	0.397 [0.379, 0.416]	21	2.779	True
pds	15029	20	SCP-T-split-N	0.909 [0.899, 0.920]	13	2.779	False
pds	15029	20	TJ+	0.924 [0.915, 0.931]	11	2.779	False
pds	15029	20	TJ+N	0.922 [0.914, 0.930]	10.6	2.779	False
pds	15029	20	Transport	0.989 [0.984, 0.993]	24.5	2.779	False
pds	15029	20	WCP-S	0.364 [0.344, 0.384]	27.9	2.779	True
sbc	15029	20	ASTM-PI	0.796 [0.793, 0.799]	5.05	1.091	True
sbc	15029	20	Gauss-LOO	0.898 [0.890, 0.906]	4.8	1.091	False
sbc	15029	20	SCP-S	0.787 [0.783, 0.790]	5.11	1.091	True
sbc	15029	20	SCP-T-split-N	0.899 [0.886, 0.911]	5.41	1.091	False
sbc	15029	20	TJ+	0.905 [0.895, 0.913]	4.91	1.091	False
sbc	15029	20	TJ+N	0.906 [0.898, 0.915]	4.99	1.091	False
sbc	15029	20	WCP-S	0.999 [0.999, 0.999]	∞	1.091	False
pds	15029	30	ASTM-PI	0.458 [0.438, 0.479]	16.1	2.398	True

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Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
pds	15029	30	Gauss-LOO	0.916 [0.907, 0.923]	8.73	2.398	False
pds	15029	30	LF-conv	0.892 [0.883, 0.900]	8.77	2.398	False
pds	15029	30	SCP-S	0.448 [0.428, 0.468]	16.5	2.398	True
pds	15029	30	SCP-T-split-N	0.934 [0.925, 0.943]	11.6	2.398	False
pds	15029	30	TJ+	0.915 [0.908, 0.922]	9.24	2.398	False
pds	15029	30	TJ+N	0.910 [0.903, 0.916]	8.94	2.398	False
pds	15029	30	Transport	0.989 [0.985, 0.993]	16.6	2.398	False
pds	15029	30	WCP-S	0.413 [0.392, 0.433]	19.3	2.398	True
sbc	15029	30	ASTM-PI	0.802 [0.800, 0.805]	4.92	1.07	True
sbc	15029	30	Gauss-LOO	0.897 [0.891, 0.903]	4.7	1.07	False
sbc	15029	30	SCP-S	0.793 [0.790, 0.795]	4.97	1.07	True
sbc	15029	30	SCP-T-split-N	0.935 [0.927, 0.943]	5.23	1.07	False
sbc	15029	30	TJ+	0.902 [0.895, 0.909]	4.75	1.07	False
sbc	15029	30	TJ+N	0.902 [0.895, 0.909]	4.84	1.07	False
sbc	15029	30	WCP-S	0.999 [0.999, 0.999]	∞	1.07	False
pds	15029	50	ASTM-PI	0.507 [0.489, 0.525]	12.4	2.032	True
pds	15029	50	Gauss-LOO	0.905 [0.900, 0.910]	7.55	2.032	False
pds	15029	50	LF-conv	0.881 [0.876, 0.885]	7.66	2.032	True
pds	15029	50	SCP-S	0.497 [0.478, 0.515]	12.7	2.032	True
pds	15029	50	SCP-T-split-N	0.917 [0.909, 0.926]	9.22	2.032	False
pds	15029	50	TJ+	0.903	7.6	2.032	False

Pipeline	Target instrument	m	Method	Coverage [95% CI]	Median IS	RMSEP	Under-covering
pds	15029	50	TJ+N	0.907 [0.898, 0.908]	7.6	2.032	False
pds	15029	50	Transport	0.983 [0.901, 0.912]	13.3	2.032	False
pds	15029	50	WCP-S	0.469 [0.978, 0.988]	13.5	2.032	True
sbc	15029	50	ASTM-PI	0.806 [0.449, 0.488]	4.84	1.057	True
sbc	15029	50	Gauss-LOO	0.906 [0.804, 0.808]	4.58	1.057	False
sbc	15029	50	SCP-S	0.796 [0.902, 0.911]	4.9	1.057	True
sbc	15029	50	SCP-T-split-N	0.923 [0.794, 0.798]	4.93	1.057	False
sbc	15029	50	TJ+	0.911 [0.916, 0.930]	4.63	1.057	False
sbc	15029	50	TJ+N	0.910 [0.906, 0.916]	4.69	1.057	False
sbc	15029	50	WCP-S	0.999 [0.905, 0.915]	∞	1.057	False
				[0.999, 0.999]			

Table S7. Scaling of computational cost with m

Table S7. Wall-clock time of a single transfer fit with prediction and of TJ+N (including five CV+ refits) versus the number of standards m (single CPU core, median of 10 runs).

dataset	kind	m	n_test	step	median_s
corn	pds	10	20	single transfer fit + prediction	0.0047
corn	pds	10	20	TJ+N (m LOO + 5 CV fits)	0.101
corn	pds	20	10	single transfer fit + prediction	0.0044
corn	pds	20	10	TJ+N (m LOO + 5 CV fits)	0.146
tablets	pds	10	450	single transfer fit + prediction	0.0349
tablets	pds	10	450	TJ+N (m LOO + 5 CV fits)	0.8028
tablets	pds	20	440	single transfer fit + prediction	0.0239

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dataset	kind	m	n_test	step	median_s
tablets	pds	20	440	TJ+N (m LOO + 5 CV fits)	1.2477
tablets	pds	30	430	single transfer fit + prediction	0.0248
tablets	pds	30	430	TJ+N (m LOO + 5 CV fits)	1.7399
tablets	pds	50	410	single transfer fit + prediction	0.0227
tablets	pds	50	410	TJ+N (m LOO + 5 CV fits)	2.56
mango	sbc	10	715	single transfer fit + prediction	0.0004
mango	sbc	10	715	TJ+N (m LOO + 5 CV fits)	0.0226
mango	sbc	20	705	single transfer fit + prediction	0.0005
mango	sbc	20	705	TJ+N (m LOO + 5 CV fits)	0.033
mango	sbc	30	695	single transfer fit + prediction	0.0004
mango	sbc	30	695	TJ+N (m LOO + 5 CV fits)	0.0425
mango	sbc	50	675	single transfer fit + prediction	0.0004
mango	sbc	50	675	TJ+N (m LOO + 5 CV fits)	0.0634

Table S8. Normalised Gaussian competitor (E9)

Table S8. TJ+N versus Gaussian leave-one-out intervals without (Gauss-LOO) and with (Gauss-LOO-N) difficulty normalisation (200 repetitions; corn and tablets PDS, mango SBC, mango instruments PDS and SBC).

Set	Method	α	Mean coverage	Under-covering	Median IS	Median width
corn	Gauss-LOO	0.05	0.945	7/24	1.27	1.05
corn	Gauss-LOO	0.1	0.903	0/24	1.07	0.863
corn	Gauss-LOO	0.2	0.818	0/24	0.878	0.658
corn	Gauss-LOO-N	0.05	0.951	0/24	1.26	1.09
corn	Gauss-LOO-N	0.1	0.905	0/24	1.07	0.897
corn	Gauss-LOO-N	0.2	0.811	0/24	0.884	0.683
corn	TJ+N	0.05	0.986	0/24	inf	1.07
corn	TJ+N	0.1	0.932	0/24	1.15	0.996
corn	TJ+N	0.2	0.835	0/24	0.907	0.732
mango	Gauss-LOO	0.05	0.955	0/4	5.13	4.33
mango	Gauss-LOO	0.1	0.902	0/4	4.45	3.59

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Set	Method	α	Mean coverage	Under-covering	Median IS	Median width
mango	Gauss-LOO	0.2	0.792	1/4	3.76	2.76
mango	Gauss-LOO-N	0.05	0.956	0/4	5.05	4.29
mango	Gauss-LOO-N	0.1	0.902	0/4	4.38	3.55
mango	Gauss-LOO-N	0.2	0.790	2/4	3.72	2.74
mango	TJ+N	0.05	0.968	0/4	5.16	4.41
mango	TJ+N	0.1	0.902	0/4	4.4	3.6
mango	TJ+N	0.2	0.803	0/4	3.75	2.83
mango_inst	Gauss-LOO	0.05	0.950	4/24	7.48	6.35
mango_inst	Gauss-LOO	0.1	0.908	0/24	6.46	5.23
mango_inst	Gauss-LOO	0.2	0.820	0/24	5.39	4.01
mango_inst	Gauss-LOO-N	0.05	0.958	6/24	7.88	6.67
mango_inst	Gauss-LOO-N	0.1	0.915	0/24	6.64	5.5
mango_inst	Gauss-LOO-N	0.2	0.810	3/24	5.44	4.22
mango_inst	TJ+N	0.05	0.974	0/24	11.9	7.73
mango_inst	TJ+N	0.1	0.919	0/24	6.97	5.74
mango_inst	TJ+N	0.2	0.827	0/24	5.48	4.22
tablets	Gauss-LOO	0.05	0.949	6/12	20.1	16.5
tablets	Gauss-LOO	0.1	0.910	0/12	17.5	13.7
tablets	Gauss-LOO	0.2	0.825	1/12	14.9	10.6
tablets	Gauss-LOO-N	0.05	0.948	2/12	21.4	16
tablets	Gauss-LOO-N	0.1	0.907	1/12	18.1	13.3
tablets	Gauss-LOO-N	0.2	0.819	0/12	14.8	10.3
tablets	TJ+N	0.05	0.971	0/12	24.7	18.4
tablets	TJ+N	0.1	0.912	0/12	18.2	13
tablets	TJ+N	0.2	0.820	0/12	14.8	10

Fig. S1. CNN training

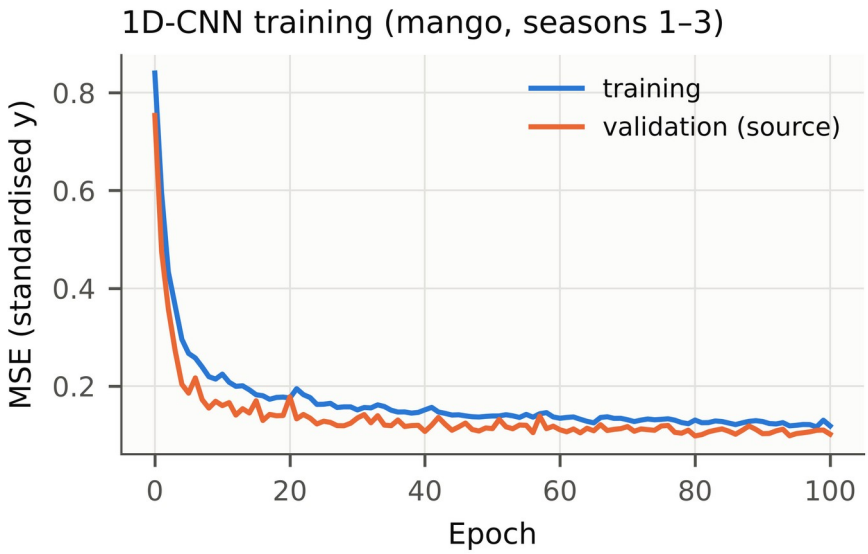


Fig. S1. Training and validation loss of the 1D-CNN (mango, seasons 1–3).

Fig. S2. Friedman ranks

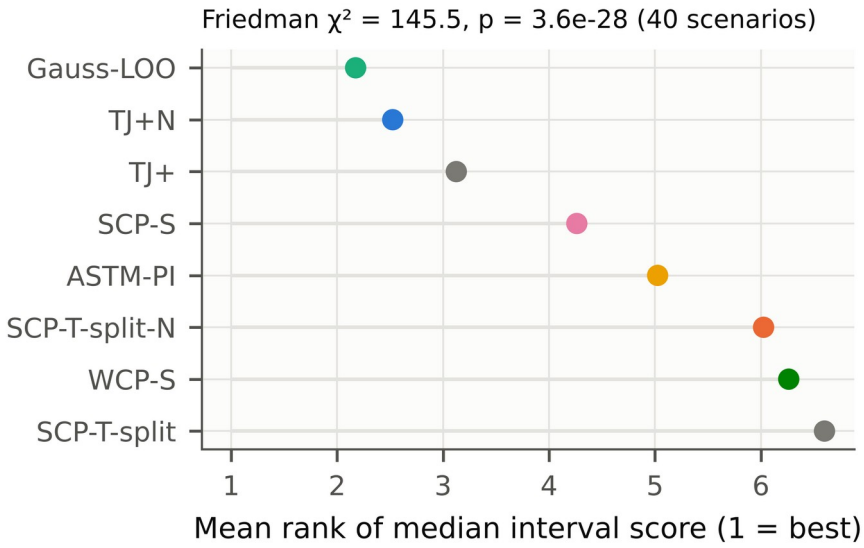


Fig. S2. Mean Friedman ranks of the scenario median interval score over the 40 primary scenarios (1 = best).

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